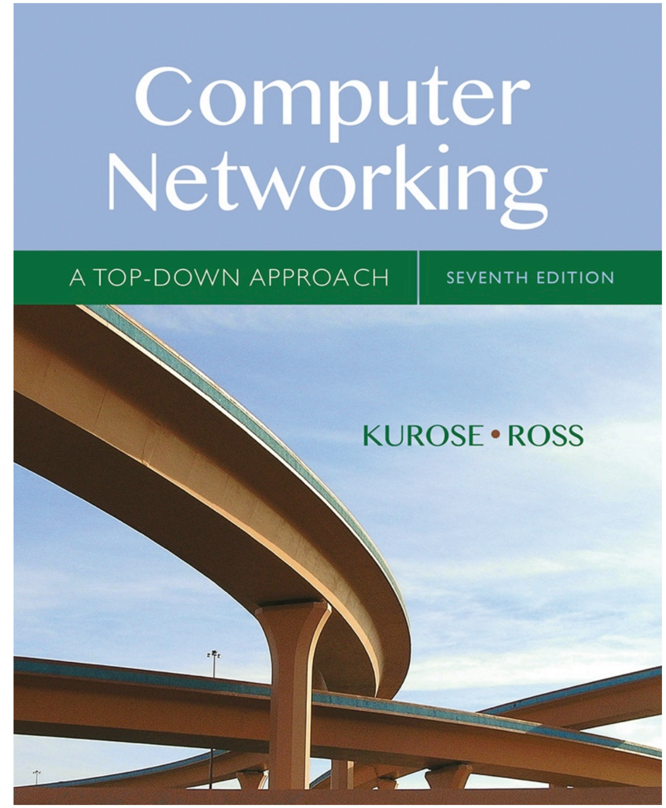
Chapter 6 The Link Layer and LANs



Computer Networking: A Top Down Approach

7th edition
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Pearson/Addison Wesley
April 2016

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Thanks and enjoy! JFK/KWR

Link layer, LANs: outline

6.1 introduction, services

6.2 error detection, correction

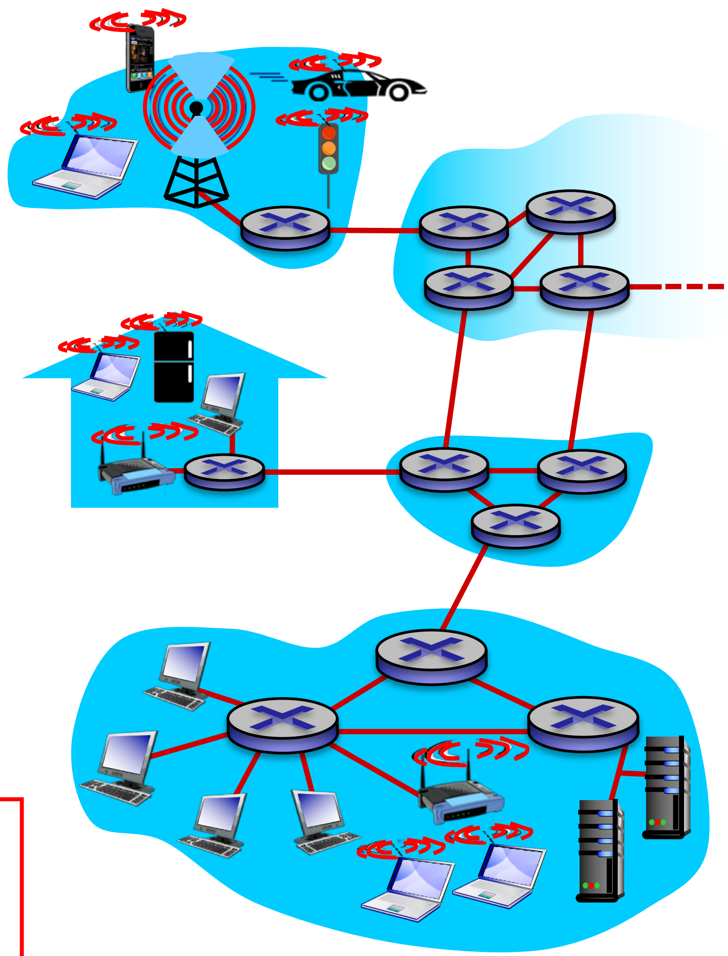
6.3 multiple access protocols

6.4 LANs
- addressing, ARP
- Ethernet

Link layer: introduction

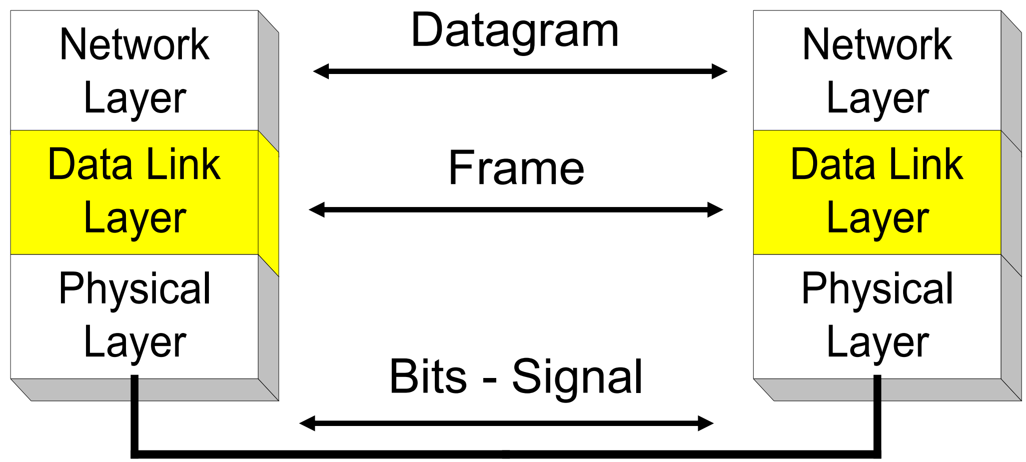
terminology:

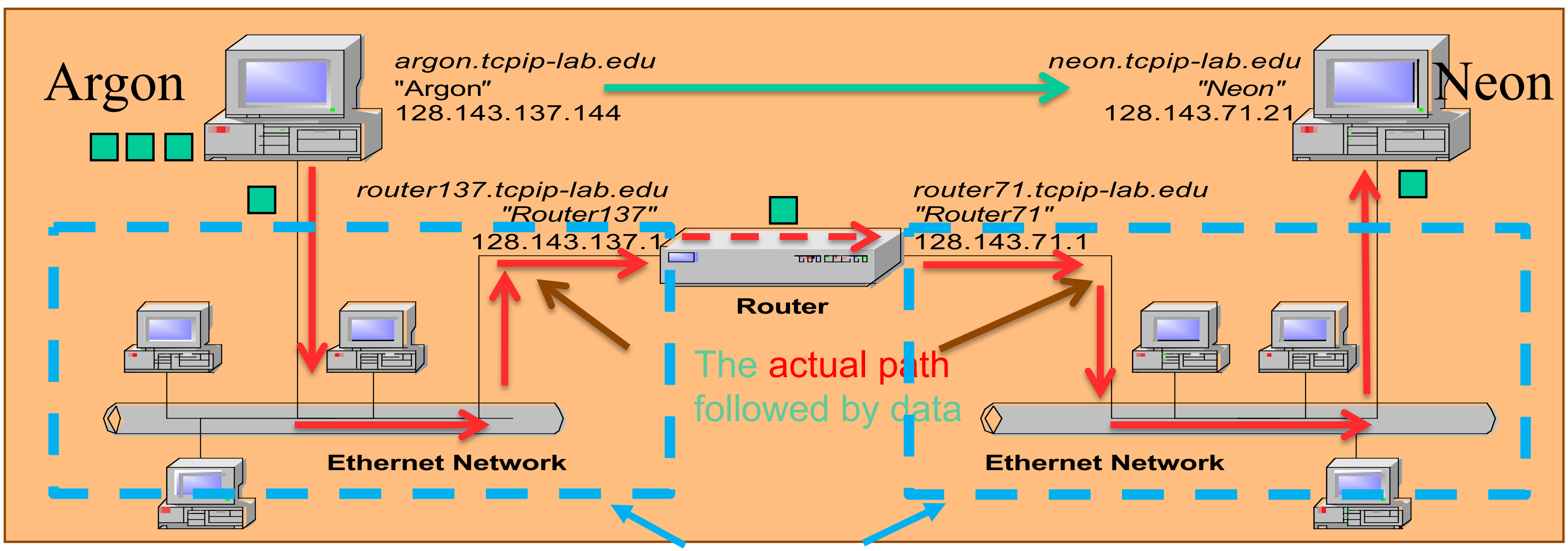
- hosts and routers: nodes
- communication channels that connect adjacent nodes along communication path: links
  - wired links
  - wireless links
  - LANs
- layer-2 packet: *frame*, encapsulates datagram



data-link layer has responsibility of transferring datagram from one node to *physically adjacent* node over a link

Data Link Layer in IETF





Link layer services

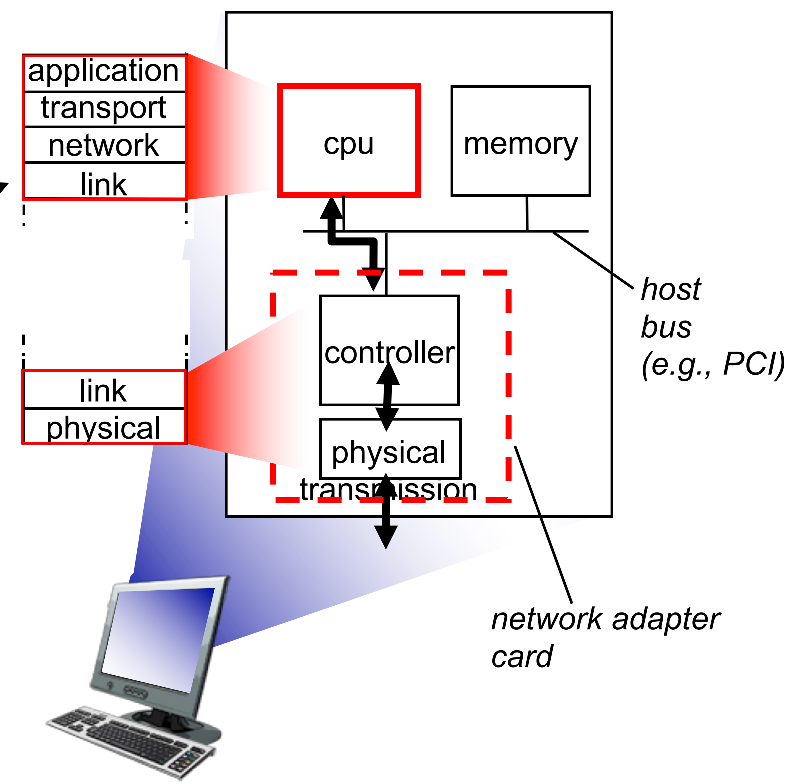
- *framing:*
  - encapsulate datagram into a frame, adding header, trailer
  - header contains hardware (network interface card) source and destination addresses
- *link access:*
  - channel access – when to transmit
    - point to point - half vs full duplex (with half duplex, nodes at both ends of link can transmit, but not at same time, full duplex allows transmission in both directions simultaneously)
    - shared medium - much more involved
- *flow control:*
  - pacing between sending and receiving nodes

Link layer services (more)

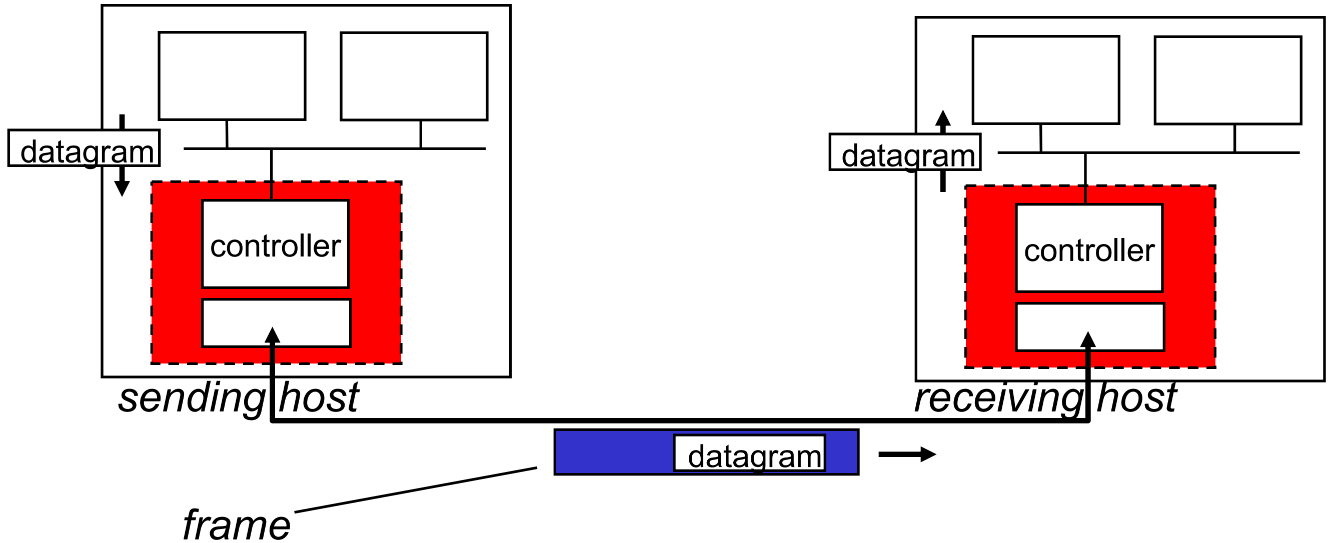
- *reliable delivery between nodes*
  - links have different error rates –> different error control schemes
    - simple schemes (e.g., detection only) used on low bit-error links (fiber, coax, ……)
    - more complex schemes (e.g., includes error recovery) used on wireless links - have high error rates

Where is the link layer implemented?

- in each and every host
- link layer implemented in "adaptor" (aka *network interface card* NIC) or on a chip
  - e.g., Ethernet card, 802.11 card; Ethernet chipset
  - implements link, physical layer
- attaches into host's system buses
- link-layer is combination of software in host CPU (addressing), hardware (framing, error recovery, access control), and firmware on NIC



Adaptors communicating



- sending side:
  - encapsulates datagram in frame
  - adds error checking bits, flow control, error recovery, etc.
- receiving side
  - looks for errors, flow control, error recovery, etc.
  - extracts datagram, passes to upper layer at receiving side

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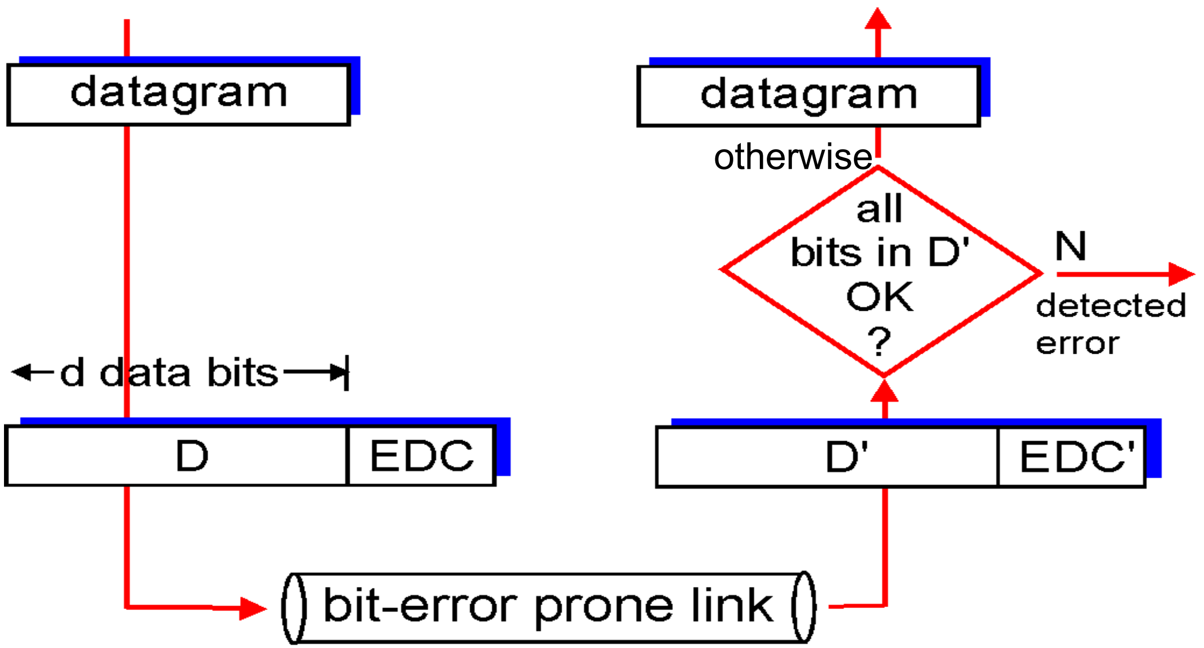
Reliable or unreliable delivery

- errors caused by signal attenuation, noise, interference
- *error detection:*
  - receiver detects presence of errors in received information
    - can drop frame and ignore error relying on higher layers
    - can try to recover
- *error recovery:*
  - receiver, after detecting errors, can try to recover the data
    - correction: receiver identifies *and corrects* the bit error(s)
    - retransmission: signals sender for retransmission of errored frame

Error detection

EDC = Error Detection and Correction bits (redundancy)
D = Data protected by error checking, may include header fields

- Error detection not 100% reliable!
  - protocol may miss some errors, but rarely
  - larger EDC field yields better detection and maybe some correction



D’≠D -> ??
What next?

Error recovery

- correction - EDC bits can also be used to correct some bit errors (in conjunction with detection)
  - correction requires larger number of EDC bits than just detection - high overhead
  - used for applications that cannot wait for correct data to be retransmitted
- retransmission - when error(s) detected
  - receiver drops packet and data is retransmitted by sender
    1. waits for sender to timeout (no ACK received) and retransmit, or
    2. receiver sends a negative ACK (NACK) (speeds up process)

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Multiple access links, protocols

two types of “links”:

- point-to-point: -> two people talking one on one
  - point-to-point link (wired or wireless) between two devices
- *broadcast (shared wire or medium): –> group conversation – has to be managed to avoid interference*
  - old-fashioned Ethernet (bus or hub)
  - upstream cable network
  - 802.11 wireless LAN (WiFi)

shared wire (e.g., cabled Ethernet)

shared RF (e.g., 802.11 WiFi)

--->

humans at a cocktail party (shared air, acoustical)

Multiple access protocols

- single *shared* channel
- if two or more simultaneous transmissions on the channel by nodes → interference
  - *interference → collision* if node receives two or more signals at the same time

multiple access protocol

- distributed algorithm that determines how nodes share channel, i.e., determine when node(s) can transmit to hopefully avoid *collision*
- communication about channel sharing often uses channel itself!
  - i.e., inband, instead of out-of-band (separate) channel for coordination

MAC protocols: taxonomy

three broad classes:

- *channel partitioning*
  - divide channel into smaller "pieces" (time slots, frequency, code)
  - allocate piece to node for exclusive use
- *random access*
  - no allocations, allow collisions
  - "recover" from collisions
- *"taking turns"*
  - nodes take turns, but nodes with more to send can take longer turns

Random access protocols

- when node has packet to send
  - transmit, possibly at full channel data rate R or in random time slots
  - no *a priori* coordination among nodes
- two or more transmitting nodes on channel or use same time slot ➜ "collision"
- random access MAC protocol should specify:
  - how to detect collisions – signal comparison, no ACK, etc
  - how to recover from collisions (e.g., via *delayed* retransmissions)
- examples of random access MAC protocols:
  - ALOHA, slotted ALOHA
  - CSMA, CSMA/CD, CSMA/CA

CSMA (carrier sense multiple access)

CSMA: listen before transmit:

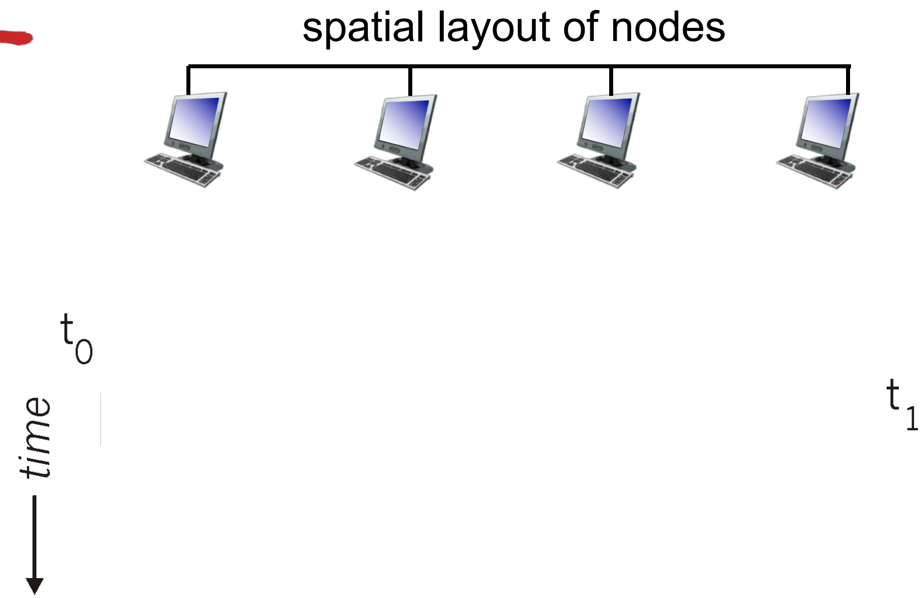
if channel sensed idle: transmit entire frame

- if channel sensed busy, defer transmission

- human analogy: don't interrupt when someone starts talking!

CSMA collisions

- collisions *can* still occur: **propagation delay** means two nodes may not hear each other's transmission
- collision: entire packet transmission time wasted
  - **distance & propagation delay** play role in in determining collision probability

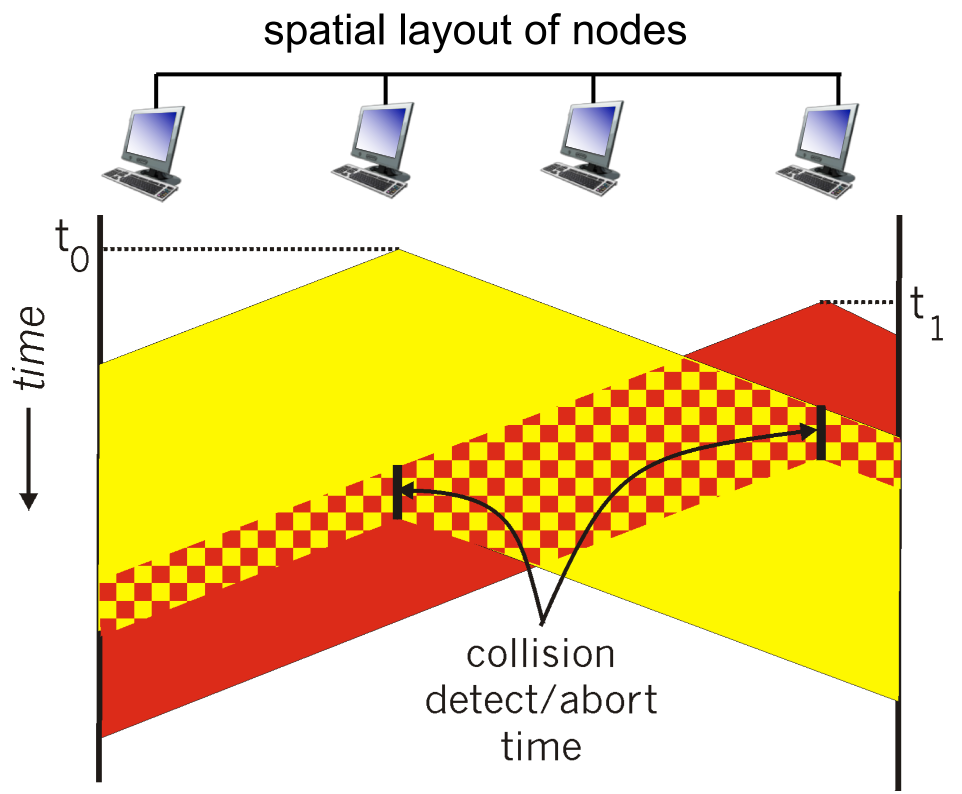


CSMA/CD (collision detection)

CSMA/CD: carrier sensing, deferral as in CSMA
- collisions *detected* within short time
- colliding transmissions aborted, reducing channel wastage

- collision detection:
  - easy in wired LANs: measure signal strengths, compare transmitted, received signals – requires a constraint on distance and min frame size
  - difficult in wireless LANs: received signal strength overwhelmed by local transmission strength
- human analogy - abort: the polite conversationalist stops when someone else starts talking.

CSMA/CD (collision detection)



CSMA/CD algorithm

1. NIC receives datagram from network layer, creates frame
2. If NIC senses channel idle, starts frame transmission. If NIC senses channel busy, waits until channel idle, then transmits (sometimes with prob. "p"). Why???
3. If NIC transmits entire frame without detecting another transmission, NIC is done with frame!
4. If NIC detects another transmission while transmitting, aborts and sends jam signal
5. After aborting, NIC enters *binary (exponential) backoff:*
   - after *m*th collision, NIC chooses *K* randomly from $\{0,1,2, \ldots, 2^m-1\}$. NIC waits K·512 bit times, returns to Step 2
   - longer backoff interval with more collisions (*m*↑)

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